Anthony M. Nguyen

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Summary and Interests

Electrical and Computer Engineering Ph.D. student at University of Michigan with machine learning and robotics research experience. Interested in control theory, linear systems, multi-agent systems, safety, reinforcement learning, distributed optimization, state estimation, robotics, and machine learning.

Skills

Relevant Coursework: Robotic Sensing and Estimation, Robotic Planning and Learning, Linear Systems, Control Theory, Optimization, Probability Theory, Random Processes, Machine Learning, Signal Processing
Programming Languages: Python (*NumPy, PyTorch, JAX, TensorFlow*), MATLAB, C, Java, Verilog, VHDL
Tools: MATLAB, QuestaSim, Active-HDL, Xilinx Vivado, LaTeX/Overleaf, RedHat Linux

Education

University of Michigan (UMich) — Ann Arbor, MI
Ph.D., Electrical and Computer Engineering (Advisor: Necmiye Ozay) — *August 2026 - Present*

University of California, San Diego (UCSD) — La Jolla, CA
B.S., Electrical Engineering (GPA: 3.982/4.000; Magna Cum Laude) — *September 2023 - June 2026*

Experience

Existential Robotics Laboratory (UCSD) — La Jolla, CA
Undergraduate Student Researcher — *January 2025 – Present*

- Developed Monte Carlo tree search (MCTS) algorithms for optimal control of partially observable Markov decision processes (POMDPs); Successfully implemented in Python for sensor scheduling problems
- Discovered, formalized, and proved relationships between different approaches to particle flow particle filtering (PFPF); Developed and proved methods to avoid numerical instability for PFPF
- Implemented PFPF methods in Python for linear and nonlinear observation models using analytical and numerical methods; Used ordinary differential equation (ODE) solvers (*SciPy, Diffrax*) and GPU-acceleration libraries (*JAX*)

L3Harris Technologies, Inc. — Carlsbad, CA
Electrical Engineering Intern — *June 2025 – September 2025*

- Studied and presented on latest generative model and robotics technologies to the team
- Migrated a communication design in Verilog from Altera to Xilinx technology; Updated designs for the company's new hardware; Recreated transmission flagging logic on the new design

Cognicom, Inc. — San Diego, CA
Electrical Engineering Intern — *January 2023 – September 2023*

- Implemented in Verilog serial communication protocols, created a precision timer circuit, and created a Xilinx Zynq bare-metal C program to control an LCD screen for a tactical radio

UC San Diego Institute of Engineering in Medicine (IEM): OPALS — La Jolla, CA
Research Intern — *June 2022 – August 2022*

- Worked to implement and analyze UCSD Biophotonics Lab research papers relating to cell microsurgery robots
- Worked with a graduate student to test results for a research project about DNA repair in cells treated with red light

Awards

Magna Cum Laude (UCSD) — June 2026

- Top 4% in GPA of all graduating students in the class of 2026 at UCSD.

ECE Booker Memorial Honors Award (UCSD) — May 2026

- Awarded to students in UCSD ECE who graduate with a GPA above 3.7.

Provost Honors (UCSD) — Fall 2023 – Spring 2026

- Awarded to at the end of each quarter to students who achieve a 3.5 or higher UC GPA in at least 12 graded units

George Eastman Young Leaders Award (High School) — May 2022

- Awarded for strong leadership, high grades and challenging courses, and extensive extracurricular activities